

2. Brief review of the parts of the RCM relevant to design.

2.1. *Part one: units, treatments, potential outcomes.* Three basic concepts are used to define causal effects in the RCM. A unit is a physical object, for example, a patient, at a particular place and point in time, say, time t . A treatment is an action or intervention that can be initiated or withheld from that unit at t (e.g., an anti-hypertensive drug, a job-training program); if the active treatment is withheld, we will say that the unit has been exposed to the control treatment. Associated with that unit are two potential outcomes at a future point in time, say, $t^* > t$: the value of some outcome measurements Y (e.g., cholesterol level, income, possibly vector valued with more than one component) if the active treatment is given at t , $Y(1)$, and the value of Y at the same future point in time if the control treatment is given at t , $Y(0)$. The causal effect of the treatment on that unit is defined to be the comparison of the treatment and control potential outcomes at t^* (e.g., their difference, their ratio, the ratio of their squares). The times t can vary from unit to unit in a population of N units, but typically the intervals, $t^* - t$, are essentially constant across the N units.

The full set of potential outcomes comprises all values of the outcome Y that could be observed in some real or hypothetical experiment comparing the active treatment to the control treatment in a population of N units. Under the “Stable Unit-Treatment Value Assumption (SUTVA)” [Rubin (1980, 1990a)], the full set of potential outcomes for two treatments and the population of N units can be represented by an array with N rows, one for each unit, and two “super” columns, one for $Y(0)$ and one for $Y(1)$, “super” in the sense that Y can be multi-component. The fundamental problem facing causal inference [Holland (1986); Rubin (1978), Section 2.4] is that for the i th unit, only one of the potential outcomes for each unit, either $Y(0)$ or $Y(1)$, can ever be observed. In contrast to outcome variables, covariates are variables, X , that for each unit take the same value no matter which treatment is applied to the unit, such as quantities determined (e.g., measured) before treatments are assigned (e.g., age, pre-treatment blood pressure or pre-treatment education). The values of all these variables under SUTVA is the N row array, $[X, Y(0), Y(1)]$, which is the object of causal inference called “the science.”

A causal effect is, by definition, a comparison of treatment and control potential outcomes on a common set of units; for example, the average $Y(1)$ minus the average $Y(0)$ across all units, or the median $\log Y(1)$ versus the median $\log Y(0)$ for those units who are female between 31 and 35 years old, as indicated by their X values, or the median $[\log Y(1) - \log Y(0)]$ for those units whose $Y(0)$ and $Y(1)$ values are both positive. It is critically important in practice to keep this definition firmly in mind.

This first part of the RCM is conceptual and can, and typically should, be conducted before seeing any data, especially before seeing any outcome data. It forces the conceptualization of causal questions in terms of real or hypothetical manipulations: “No causation without manipulation” [Rubin (1975)]. The formal use of potential outcomes to define unit-level causal effects is due to Neyman in 1923 [Rubin (1990a)] in the context of randomized experiments, and was a marvelously clarifying contribution. But evidently this notation was not formally extended to nonrandomized settings until Rubin (1974), as discussed in Rubin (1990a, 2005) and Imbens and Rubin (2008a, 2008b).

The intuitive idea behind the use of potential outcomes to define causal effects must be very old. Nevertheless, in the context of nonrandomized observational studies, prior to 1974 everyone appeared to use the “observed outcome” notation when discussing “formal” causal inference. More explicitly, letting W be the column vector indicating the treatment assignments for the units ($W_i = 1$ if treated, $W_i = 0$ if control), the observed outcome notation replaces the array of potential outcomes $[Y(0), Y(1)]$ with Y_{obs} , where the i th component of Y_{obs} is

$$(2.1) \quad Y_{\text{obs},i} = W_i Y_i(1) + (1 - W_i) Y_i(0).$$

The observed outcome notation is inadequate in general, and can lead to serious errors—see, for example, the discussions in Holland and Rubin (1983) on Lord’s paradox, and in Rubin (2005), where errors are explicated that Fisher made because (I believe) he eschewed the potential outcome notation. The essential problem with Y_{obs} is that it mixes up the science [i.e., $Y(0)$ and $Y(1)$] with what is done to learn about the science via the assignment of treatment conditions to the units (i.e., W_i).

2.2. Part 2: the assignment mechanism. The second part of the RCM is the formulation, or positing, of an assignment mechanism, which describes the reasons for the missing and observed values of $Y(0)$ and $Y(1)$ through a probability model for W given the science:

$$(2.2) \quad \Pr(W|X, Y(0), Y(1)).$$

Although this general formulation, with the possible dependence of assignments on the yet to be observed potential outcomes, arose first in Rubin (1975), special cases were much discussed prior to that. For example, randomized experiments [Neyman (1923, 1990), Fisher (1925)] are “unconfounded” [Rubin (1990b)],

$$(2.3) \quad \Pr(W|X, Y(0), Y(1)) = \Pr(W|X),$$

and they are “probabilistic” in the sense that their unit level probabilities, or propensity scores e_i , are bounded between 0 and 1:

$$(2.4) \quad 0 < e_i < 1,$$

where

$$(2.5) \quad e_i \equiv \Pr(W_i = 1 | X_i).$$

When the assignment mechanism is both probabilistic [(2.4) and (2.5)] and unconfounded (2.3), then for all assignments W that have positive probability, the assignment mechanism generally can be written as proportional to the product of the unit level propensity scores, which emphasizes the importance of propensity scores in design:

$$(2.6) \quad \Pr(W | X, Y(0), Y(1)) \propto \prod_{i=1}^N e_i \quad \text{or} \quad = 0.$$

The collection of propensity scores defined by (2.5) is the most basic ingredient of an unconfounded assignment mechanism because of (2.6), and its use for objectively designing observational studies will be developed and illustrated here, primarily in Section 4, but also in the context of a more complex design discussed in Section 5.

The term “propensity scores” was coined in Rosenbaum and Rubin (1983), where an assignment mechanism satisfying (2.4) and (2.5) is called “strongly ignorable,” a stronger version of “ignorable” assignment mechanisms, coined in Rubin (1976a, 1978), which allows possible dependence on observed values of the potential outcomes, Y_{obs} defined by (2.1), such as in a sequential experiment:

$$\Pr(W | X, Y(0), Y(1)) = \Pr(W | X, Y_{\text{obs}}).$$

But until Rubin (1975), randomized experiments were not defined using (2.3) and (2.4), which explicitly show such experiments’ freedom from any dependence on observed or missing potential outcomes. Instead, randomized experiments were *described* in such a way that the assignments only depended on available covariates, and so implicitly did not involve the potential outcomes themselves. But explicit mathematical notation, like Neyman’s, can be a major advance over implicit descriptions.

Other special versions of assignment mechanisms were also discussed prior to Rubin (1975, 1978), but without the benefit of explicit equations for the assignment mechanism showing possible dependence on the potential outcomes. For example, in economics, Roy (1951) described, without equations or notation, “self-optimizing” behavior where each unit chooses the treatment with the optimal outcome. And another well-known example from economics is Haavelmo’s (1944) formulation of supply and demand behavior. But these and other formulations in economics and elsewhere did not use the notation of an assignment mechanism, nor did they have methods of statistical inference for causal effects based on the assignment mechanism. Instead, “regression” models were used to predict $Y_{\text{obs},i}$ from X_i and W_i ,

with possible restrictions on some regression coefficients and/or on “error” terms. In these models certain regression coefficients (e.g., for W_i or for interactions with W_i) were interpreted as causal effects; analogous approaches were used in other social sciences, as well as in epidemiology and medical research, and are still common. Such regression models were and are based on combined assumptions about the assignment mechanism and about the science, which were typically only vaguely explicated because they often were stated through restrictions on error terms, and therefore could, and sometimes did, lead to mistakes.

Inferential methods based only on the assumption of a randomized assignment mechanism were proposed by Fisher (1925) and described by Neyman (1923) and further developed by others [see Rubin (1990a) for some references]. The existence of these assignment-based methods, and their success in practice, documents that the model for the assignment mechanism is more fundamental for inference for causal effects than a model for the science. These methods lead to concepts such as unbiased estimation and asymptotic confidence intervals (due to Neyman), and p -values or significance levels for sharp null hypotheses (due to Fisher), all defined by the distribution of statistics (e.g., the difference of treatment and control sample means) induced by the assignment mechanism. In some contexts, such as the U.S. Food and Drug Administration’s approval of a new drug, such assignment mechanism-based analyses are considered the gold standard for confirmatory inferences.

The third and final part of the RCM is optional; it involves specifying a full probability model for the science, the quantity being conditioned on in the assignment mechanism (2.2), and therefore treated as fixed in assignment-based approaches. This approach is Bayesian, and was developed by Rubin (1975, 1978) and further developed, for example, in Imbens and Rubin (1997) and in many other places. This can, in special simple cases, lead to the use of standard models, such as ordinary least squares regression models, but such models are generally not relevant to the design of observational studies.

Of course, there are other frameworks for causal inference besides mine, including ones where models have some relevance, but that is not the topic or focus of this article. The reader interested in various uses of models on the science $(X, Y(0), Y(1))$ can examine the text by Morgan and Winship (2007), which provides a fairly comprehensive discussion of different approaches. Also informative, but with an applied and prescriptive attitude, including some advice on design issues, is the text by Shadish, Cook and Campbell (2002).